

LANDSCAPE REPORT

The Autonomous Testing Platforms Landscape, Q3 2025

Forrester's Overview Of 31 Vendors

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Summary

You can use autonomous testing platforms (ATPs) to accelerate time to value through AI-driven test automation, reduce strategic risk, increase the quality of governance, and democratize testing and cross-team collaboration. But to realize these benefits, you'll first have to select from a diverse set of vendors that vary by size, type of offering, geography, and use case differentiation. Product and application testers should use this report to understand the value they can expect from an ATP vendor, learn how vendors differ, and investigate options based on size and market focus.

Market Definition

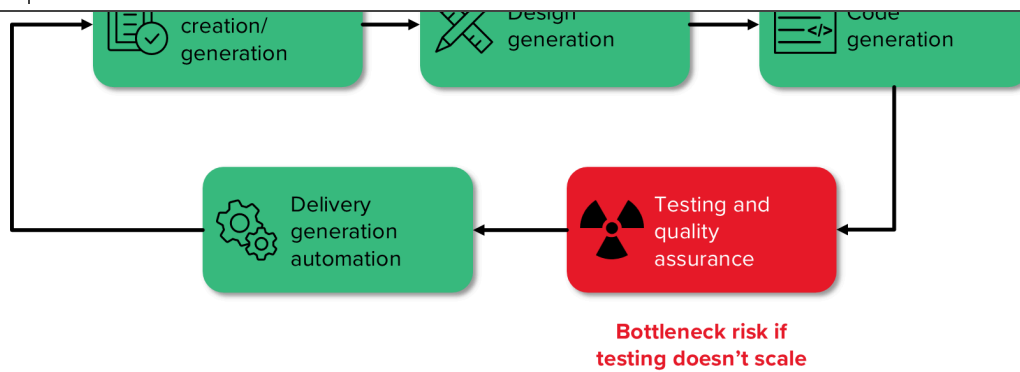
GenAI is collapsing the software development lifecycle (SDLC) with automation and generation of software assets in each stage. Testing isn't immune to this trend. Testing TuringBots, which are AI, genAI, and increasingly agentic software testing tools, are emerging from the existing continuous automation testing (CAT) platforms market as well as the emerging AI market. These solutions are creating the basis for true autonomous testing. Forrester defines ATPs as:

Platforms that combine traditional automation with AI and genAI agents to continuously perform increasingly autonomous testing tasks. These platforms can generate and execute a wide range of functional and nonfunctional end-to-end tests across products and applications — including those infused with AI — ensuring comprehensive and adaptive quality validation.

As customers increase the use of TuringBots to automate and generate requirements, designs, code, and other software assets in the SDLC, testers and testing leaders are under pressure to maintain the quality of delivery. If they don't, testing will become the bottleneck of the SDLC (see Figure 1).

FIGURE 1

Testing Risks Being The Bottleneck Of The SDLC



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Business Value

Enterprises need to address the challenges posed by the accelerated pace of modern software delivery, which is evolving from a continuous delivery model to include even more generation capabilities and autonomy. Product and application testers use ATPs to:

- **Accelerate time to value through AI-driven test automation.** ATPs reduce the time required to design, generate, and maintain test cases by leveraging AI to automate traditionally manual tasks. This acceleration is especially critical in today's compressed development cycles, where the quality-velocity paradox challenges teams to maintain high quality while releasing software faster. By automating test generation from requirements, enabling self-healing for brittle tests, and optimizing execution, these platforms help organizations ship faster without sacrificing quality for competitive advantage.
- **Reduce strategic risk and increased quality of governance.** AI-powered platforms enable risk-based testing orchestration, intelligent test scoping, and real-time analytics, allowing organizations to focus validation efforts where they matter most. Rather than treat all tests equally, autonomous platforms prioritize by business impact, recent changes, and historical defect patterns. This ensures that critical paths are always tested, reducing the risk of escaped defects and costly production issues. For regulated industries, these platforms also support governance, traceability, and compliance, making them essential for enterprise-grade QA.
- **Democratize testing and improve cross-team collaboration.** With no-code/low-code interfaces, natural language test authoring, and AI-assisted test creation, autonomous platforms empower nontechnical users to contribute to quality efforts. This democratization breaks down silos among QA, development, and business teams. Business subject-matter experts can validate user stories, product managers can define acceptance criteria, and testers can focus on strategic oversight. The result is broader test coverage, better alignment with business goals, and a more collaborative agile development process.

Market Maturity

are infused with it. Today, ATPs are an established market that:

- **Continues to evolve rapidly.** While many vendors claim AI-native or genAI-first capabilities, the reality is that the market is still in transition. There's a clear shift from traditional scripting and behavioral-driven dev (BDD) to AI-driven, agentic, and intent-based testing. Traditional testing vendors and new startups are investing significant amounts of dollars to play in this space. However, the presence of commoditized features (e.g., record/playback, UI testing, BDD) and the need to distinguish real AI from marketing hype suggest that many platforms are still maturing. Buyers are struggling to evaluate true innovation versus superficial AI claims.
- **Standardizes core capabilities while differentiating upstream.** Capabilities like DevOps integration, cloud device labs, UI recording and replay, and basic scripting are now table stakes. Differentiation is moving toward higher-order features such as test cases expressed through natural language intent specification, AI-agentic testing, risk-based orchestration, and business outcome validation. This signals a maturing baseline where vendors must now compete on strategic value, not just technical features.
- **Has buyer challenges that signal the market is still in midtransformation.** End-user organizations face challenges typical of a market in flux: continuously fast-paced innovation, fragmented toolchains, skill gaps, unclear ROI, and organizational resistance. These issues indicate that the technology is advancing quickly but operationalizing it at scale remains difficult. The need for governance, monitoring, explainability, and cross-team alignment further underscores that the market is not yet fully mature, especially for enterprisewide adoption.

FIGURE 2

Autonomous Testing Platforms Market Maturity And Key Dynamics



Key market dynamics

MAIN TREND	PRIMARY CHALLENGE	TOP DISRUPTOR
AI-driven auto testing delivers rapid innovation, adaptive automation, business alignment, and scalable continuous quality.	Buyers struggle with quality balancing, understanding of AI’s value in testing, trust, and large skill gaps for a new tester role.	Agentic AI enables self-updating, self-healing, and continuous testing with human orchestration.

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Market Dynamics

Product and application testers will face a plethora of ATP options and should pay attention to the following market dynamics. As AI platforms mature and agentic capabilities emerge, the testing landscape is shifting from scripted automation to intelligent adaptive systems. Understanding these dynamics is essential for making informed platform decisions.

- **Main trend.** AI-driven autonomous testing delivers rapid innovation, adaptive automation, business alignment, and scalable continuous quality. Platforms are evolving from script-based tools into AI systems that generate, maintain, and execute tests autonomously. This shift enables faster releases, broader business and technical test coverage, and reduced maintenance overhead. AI also empowers nontechnical users to contribute, democratizing QA. The result is a more collaborative business-aligned approach to testing.
- **Primary challenge.** Buyers struggle with quality balancing, understanding of AI’s value in testing, trust, and large skill gaps for a new tester role. Many organizations face confusion about vendor claims versus real AI capabilities. Fragmented toolchains, concerns with IP protection, hallucinations, and lack of integration hinder adoption. Testers must evolve into strategic orchestrators and AI supervisors who interpret what good quality looks like. However, most teams lack the skills and frameworks to support this shift. ROI proof and alignment of testing with business outcomes remain persistent hurdles.
- **Top disruptor.** Agentic AI enables self-updating, self-healing, and continuous testing with human orchestration. Autonomous agents can independently discover, generate, and execute tests, adapting to application changes in real time. Agents reduce test flakiness, accelerate feedback loops, and

and continuous quality at scale.

Notable Vendors

Product and application testers can start investigating specific vendors by their geographic focus, industry focus, deployment options, and size. Across all markets, Forrester defines large vendors as having \$250 million or more, medium vendors as having \$100 million to less than \$250 million, and small vendors as having less than \$100 million in category revenue (see Figure 3).

FIGURE 3

The Autonomous Testing Platforms Landscape, Q3 2025

Vendor	Geographic focus	Industry focus	Deployment model(s)	Size
ACCELQ	NA; APAC	Financial services Healthcare Retail	Hosted, private SaaS Multitenant SaaS On-premises	■
Applitools	NA; EMEA	Financial services IT/tech services Retail	Hosted, private SaaS Multitenant SaaS On-premises	■
Appvance ¹	NA; EMEA	Financial services Healthcare Insurance	Hosted, private SaaS On-premises	■
AutomatePro	NA; EMEA	Financial services IT/tech services Pharmaceuticals and medical equipment	Hosted, private SaaS On-premises	■
BrowserStack ¹	NA; EMEA	Entertainment, leisure, and hospitality Financial services Media	Multitenant SaaS	■ ■ ■
Copado ¹	NA; EMEA	Financial services Healthcare Retail	Multitenant SaaS	■
Functionize	NA; EMEA	Financial services Insurance IT/tech services	Hosted, private SaaS Multitenant SaaS	■
Katalon	NA; EMEA; APAC	Financial services Insurance IT/tech services	Hosted, private SaaS Multitenant SaaS On-premises	■
Keysight Technologies ¹	NA; EMEA	Financial services Government Manufacturing/production of high-tech products	Hosted, private SaaS Multitenant SaaS On-premises	■
LambdaTest	NA; EMEA	Financial services Insurance IT/tech services	Hosted, private SaaS Multitenant SaaS On-premises	■
Leapwork	NA; EMEA	Healthcare Pharmaceuticals and medical equipment Retail	Hosted, private SaaS On-premises	■
LogiGear ¹	NA; APAC	Financial services Gaming Manufacturing/production of high-tech products	Hybrid Multitenant SaaS	■
mabl	NA	Financial services Healthcare Retail	Multitenant SaaS	■
OpenText ¹	NA; EMEA; APAC; LATAM	Financial services Healthcare Insurance	Multitenant SaaS On-premises	■ ■ ■
Opkey	NA; EMEA	Financial services Healthcare IT/tech services	Hosted, private SaaS Multitenant SaaS On-premises	■
Panaya ¹	EMEA	IT/tech services Manufacturing/production of	Hosted, private SaaS Multitenant SaaS	■

Parasoft ¹	NA; EMEA; APAC	Financial services Manufacturing/production of high-tech products Transportation	Hosted, private SaaS On-premises	■
Perforce Software ¹	NA; EMEA; APAC	Financial services Healthcare Retail	Hosted, private SaaS Multitenant SaaS	■
PractiTest	NA; EMEA	Financial services Insurance IT/tech services	Hosted, private SaaS Multitenant SaaS	■
ProdPerfect ¹	US; EMEA	Financial services Healthcare	Multitenant SaaS	■
Progress ¹	NA; EMEA	Financial services Government Healthcare	On-premises	■
Provar	NA; EMEA	Financial services Government IT/tech services	On-premises	■
Qyru ¹	NA; EMEA	Financial services Healthcare Manufacturing/production of consumer products	Hosted, private SaaS Multitenant SaaS	■
Sauce Labs ¹	NA; EMEA	Financial services Media Retail	Hosted, private SaaS Multitenant SaaS	■ ■
Sembi	NA; EMEA	Financial services IT/tech services Retail	Multitenant SaaS On-premises	■ ■ ■
SmartBear	NA; EMEA	Financial services Healthcare IT/tech services	Hosted, private SaaS Multitenant SaaS On-premises	■ ■
Testsigma	NA; EMEA; APAC	Financial services Insurance IT/tech services	Hosted, private SaaS Multitenant SaaS On-premises	■
Tricentis	NA; EMEA	Financial services Insurance Manufacturing/production of industrial products	Multitenant SaaS On-premises	■ ■ ■
UiPath	NA; EMEA; APAC	Financial services Healthcare Manufacturing/production of industrial products	Hosted, private SaaS Multitenant SaaS On-premises	■ ■
Virtuoso	NA	Insurance IT/tech services Professional services	Hosted, private SaaS Multitenant SaaS	■
Worksoft ¹	NA; EMEA	IT/tech services Manufacturing/production of consumer products Utilities	Hosted, private SaaS On-premises	■

Size ■ ■ ■ Large ≥\$250M

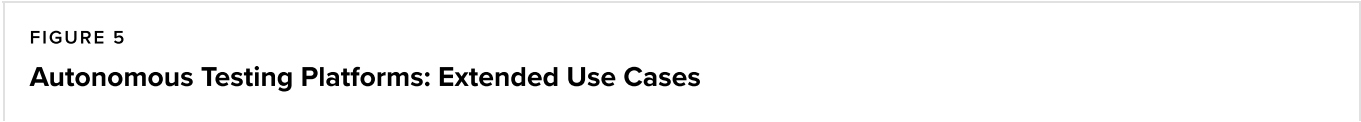
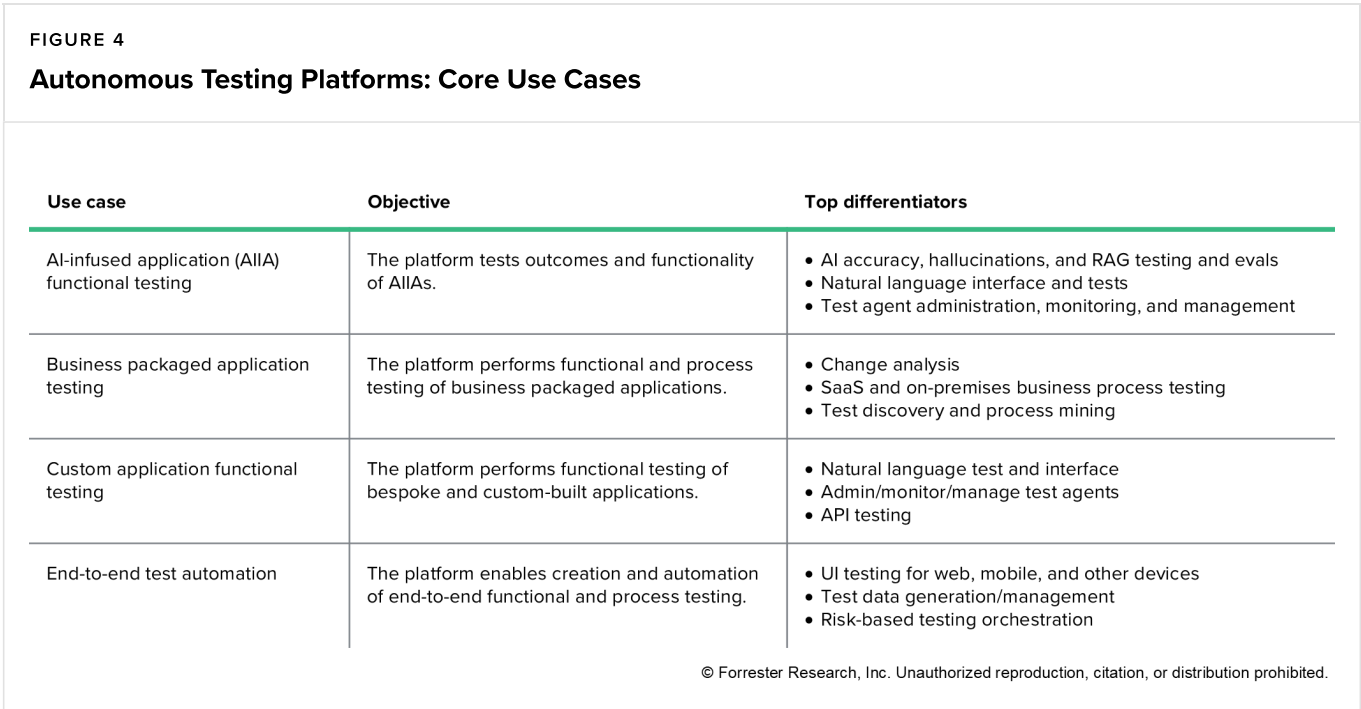
■ ■ Medium \$100M to <\$250M

■ Small <\$100M

Note: Geographic focus indicates regions where the vendor's product revenue in this category is greater than or equal to 15% of its total product revenue.

¹ The information about this vendor includes Forrester's estimates

We’ve identified the following core use cases for this market: AI-infused application functional testing, business packaged application testing, custom application functional testing, and end-to-end test automation. These are the use cases that buyers most frequently seek and that they expect ATP vendors to address (see Figure 4). Beyond these core use cases, buyers often look for vendors that focus on certain extended use cases. We’ve identified the following use cases as extended: nonfunctional performance testing, test management, testing of AI and genAI, testing of IoT and embedded testing, and visual testing (see Figure 5). Some buyers look to address these use cases in addition to the core ones, but ATP solutions may less commonly address them.



Nonfunctional performance testing	Evaluate system scalability, reliability, speed, and concurrency under varying conditions, ensuring the optimal user experience.	• Performance and benchmark testing • Service virtualization testing • API testing
Test management	Help testers orchestrate AI-driven test creation, execution, maintenance, collaboration, and analytics seamlessly.	• Test data generation/management • Natural language interface and tests • Test data and insights for enterprise/team reporting
Testing of AI and genAI	Ensure accuracy, fairness, and compliance, and limit hallucinations while mitigating risks, building trust, and driving business value.	• AI accuracy, hallucinations, and RAG testing and evals • Performance and benchmark testing • Test data generation/management
Testing of IoT and embedded testing	Validate functional and nonfunctional quality of interconnected devices and systems across hardware and software environments.	• UI testing for web, mobile, and other devices • Test data and insights for enterprise/team reporting • API testing
Visual testing	Automate seamless user experiences by validating UI accuracy, responsiveness, and consistency.	• UI testing for web, mobile, and other devices • Cloud browser and device lab testing grids • Natural language interface and tests

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Functionality By Use Case

Select the use cases that are most relevant to your business requirements and then use the following tables as a guide to choose the functionalities or capabilities that matter most for your technology evaluation and vendor selection criteria (see Figures 6 and 7).

FIGURE 6
Autonomous Testing Platforms: Functionality By Core Use Case



FIGURE 7

Autonomous Testing Platforms: Functionality By Extended Use Case

Functionality	performance testing	Test management	Testing of AI and genAI	Testing of IoT and embedded testing	Visual testing
UI testing for web, mobile, and other devices	●	●	●	●	●
Model-based, keyword-driven BDD test case design	○	○	○	●	●
Test case scripting and/or coding	●	●	●	●	○
Change analysis	○	○	●	●	●
Natural language tests and interface	●	●	●	○	○
Test agent creation and orchestration	●	●	●	○	○
Test agent administration, monitoring, and management	●	●	●	○	○
AI accuracy, hallucinations, RAG testing and evals	●	○	●	●	●
Agile planning and DevOps platforms integration	●	●	●	●	●
Cloud, browser, and device lab testing grids	●	○	●	●	●
Performance and benchmark testing	○	●	○	●	●
Test data and insights for enterprise/team reporting	○	●	●	●	●
API testing	●	●	●	○	○
Test data generation/management	●	●	●	○	●
SaaS and on-premises business process testing	●	○	○	●	●

● Primary functionality required for a given use case
○ Secondary functionality required for a given use case
● Little to no functionality required for a given use case

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Vendor Focus: Top Three Extended Use Cases

We asked each participating vendor in the report to select the top three extended use cases that it focuses on. These are three use cases, beyond the core ones, that the vendor wants customers to recognize as its areas of focus (see Figure 8). This table doesn't represent available functionalities and may not represent the only use cases that vendors serve.



Vendor	performance testing	Test management	Testing of AI and genAI	Testing of IoT and embedded testing	Visual testing
ACCELOQ		Q	Q		Q
Applitools		Q			Q
Appvance	Q	Q			Q
AutomatePro		Q	Q		
Functionize		Q			Q
Katalon		Q			Q
Keysight Technologies		Q	Q		Q
LambdaTest	Q	Q			Q
Leapwork	Q		Q		Q
mabl	Q	Q			Q
OpenText	Q	Q			Q
Opkey		Q			Q
Panaya		Q	Q		Q
Parasoft	Q	Q		Q	
Perforce Software	Q	Q			Q
PractiTest		Q		Q	
Progress	Q	Q			Q
Provar	Q	Q	Q		
Qyrus	Q		Q		Q
Sauce Labs	Q	Q			Q
Sembi		Q			Q
SmartBear	Q	Q		Q	
Testsigma		Q			Q
Tricentis	Q	Q			Q
UiPath	Q	Q	Q		
Virtuoso		Q			Q

Note: The following vendors declined to provide use case information in our questionnaire: BrowserStack, Copado, LogiGear, and ProdPerfect.

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How To Work With These Vendors

Many vendors in this landscape have a longstanding presence and have continuously evolved their platforms in response to changing technologies and market demands. Whether you're evaluating your current testing partner or exploring new options, ensure their roadmap aligns with the core and extended use cases outlined in this report. At the same time, consider experimenting with newer AI-native entrants to understand their advantages and limitations. The open question remains: Which companies will ultimately deliver comprehensive solutions for testing agentic AI systems?

Where The Market Is Going Next

The autonomous testing market might soon be approaching a pivotal inflection point. While current platforms are maturing around AI-augmented automation, the emergence of agentic AI is poised to redefine the category altogether. These systems — capable of independently discovering, generating, executing, and maintaining tests — are not just enhancing existing workflows; they are reshaping the very nature of software QA. As agentic AI becomes more capable, the market may bifurcate.

- **One path for traditional testing vendors will continuously evolve.** On this path, traditional testing vendors will continue to develop, integrating agentic capabilities into their platforms to support increasingly autonomous, continuous testing pipelines.
- **The other path will give rise to a new category of testing tools.** This new category will focus on testing AI itself — especially LLMs, autonomous agents, and AI-infused applications — where practices like red teaming, hallucination detection, and benchmark evaluation become essential. Companies like Giskard, ImpactQA.ai, and Qodo were not included in this vendor landscape because their revenue was less than our threshold and they didn't meet the extensive criteria. However, they offer eval benchmark testing, red teaming, and other new practices that might become mainstream in the future.

This signals that the market may not mature but instead may expand into a new adjacent space — one focused on AI assurance. Here, testing is no longer just about validating software functionality; it's about ensuring the trustworthiness, safety, and compliance of AI systems themselves. In short, agentic AI may catalyze the emergence of a new market for testing AI with AI.

Supplemental Material

Methodology

To complete our review, Forrester requested information from vendors. If vendors didn't share this information with us, we made estimates based on available secondary information. We've marked all estimates with a note. Forrester shared a preview of this report with participating vendors before publishing.

ACCELQ

Applitools

Appvance

AutomatePro

BrowserStack

Copado

Functionize

Katalon

Keysight Technologies

LambdaTest

Leapwork

LogiGear

mabl

OpenText

Opkey

Panaya

Parasoft

Perforce Software

PractiTest



Progress

Provar

Qyrus

Sauce Labs

Sembi

SmartBear

Testsigma

Tricentis

UiPath

Virtuoso

Worksoft

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